Wireless Channel Fingerprinting and Charting

Ray-Tracing Dataset Generation, Localization, and Dimensionality Reduction

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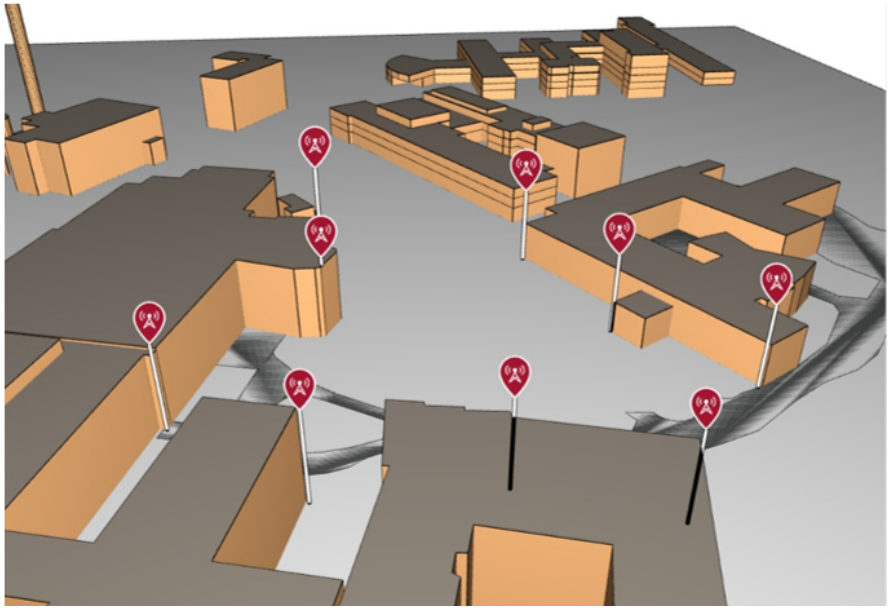
Urban Radio Propagation

- Urban environments exhibit rich multipath propagation
- Dominant mechanisms:
  - Specular reflection from building facades
  - Diffraction at building edges
  - Transmission through exterior walls
- Key channel descriptors:
  - RMS delay spread
  - Angular spread of arrival/departure
  - Aggregate path loss



Ray-Tracing for Channel Modelling

- Deterministic propagation modelling technique
- Simulates geometric and electromagnetic interactions of rays with surfaces
- Advantages over stochastic models:
  - Directly exploits site-specific scene information
  - Produces physically realistic channel characteristics
  - Enables site-dependent analysis

Dataset Generation:

- MATLAB's Ray Tracing Toolbox
- OpenStreetMap-derived 3-D model
- Configurable parameters: reflections, diffractions, frequency

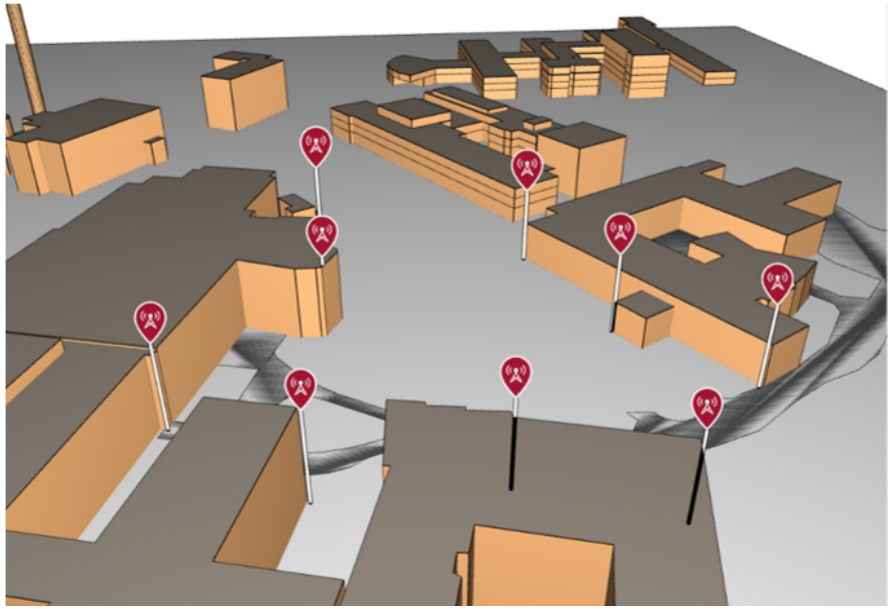
Otaniemi Campus Setup

Scene Geometry:

- Location: Espoo, Finland
- Area: 132×147 meters
- Building model from OpenStreetMap
- Material properties: concrete (walls), metal (roofs), loam (ground)

Radio Configuration:

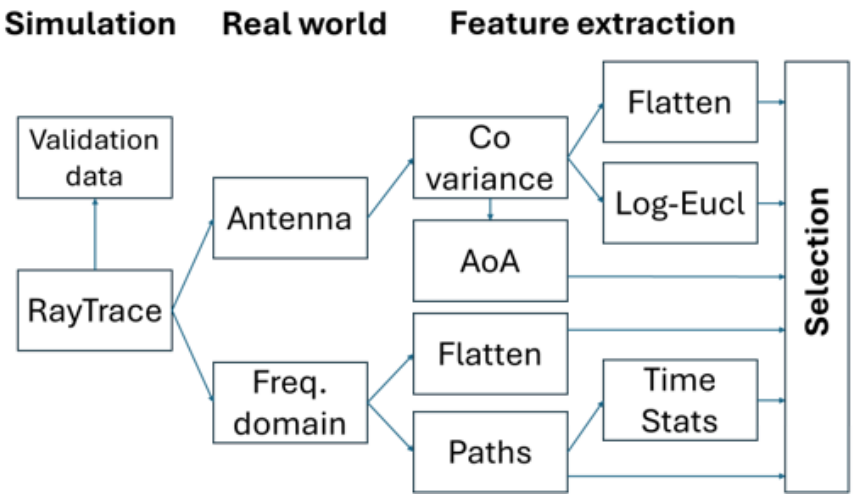
- Frequency: 3.6 GHz (5G NR Band n78)
- 9 rooftop base stations
- 16-element 4×4 antenna arrays per BS
- UE grid: 2.5 m spacing



MATLAB Ray-Tracing Dataset

Feature Extraction:

- Ray tracing parameters:
  - Max reflections: 4
  - Max diffractions: 2
  - Retained paths per link: 40 strongest
- Two feature branches:
  - Antenna-domain features (AoA, covariance)
  - Frequency-domain features (CSI)



Final Feature Set (63 dimensions):

- Angle-of-Arrival (AoA)
- Received Signal Strength (RSS)
- Path Delay

Localization Methods

- **Non-parametric:** wKNN (IDW) with $k = 3$ and Minkowski $p = 3$
- **Neural Regressors:**
  - MLP: fully-connected, 4 hidden layers
  - CNN: 1-D/2-D convolutional with cross-BS convolutions
- **Ensemble:** Average of 5 independently-initialised MLPs
- **Classification:** MLP mapping to 2.5 m grid cells (fails due to sparsity)

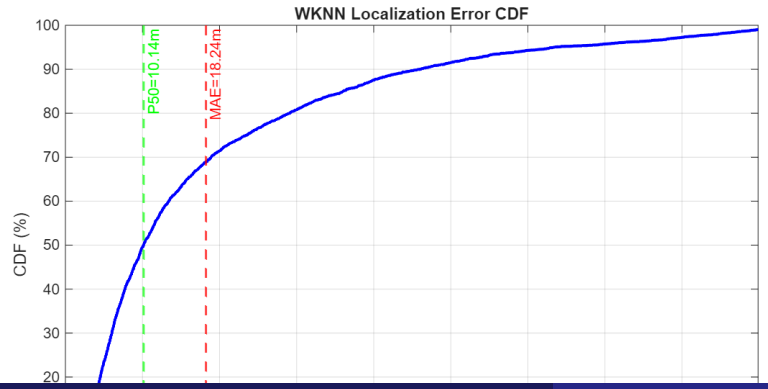
Evaluation Metrics:

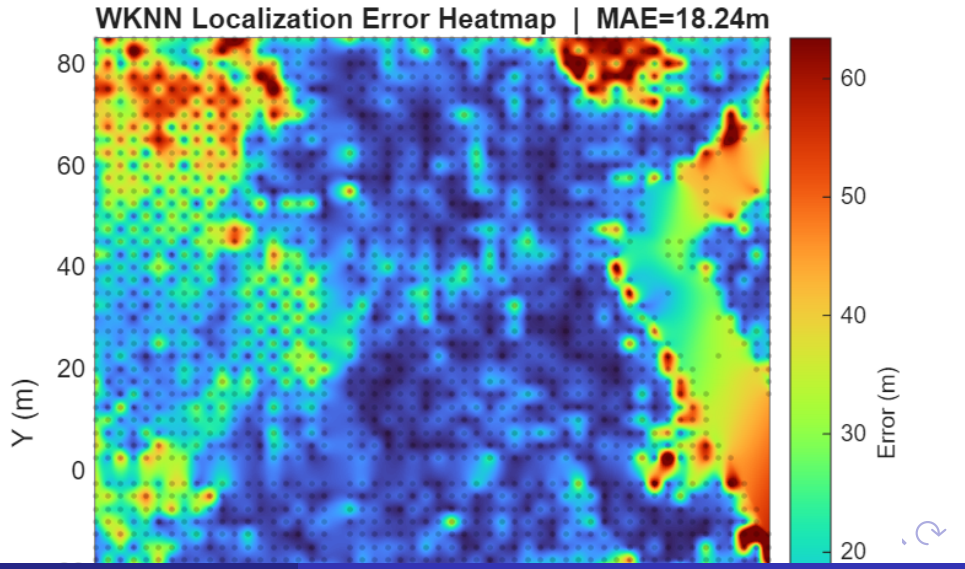
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Error percentiles (P90, P95)

Localization Results

Table: MATLAB Dataset Performance ($N_{\text{test}} = 921$, $D_f = 63$)

Method	MAE (m)	RMSE (m)	Median (m)	P90 (m)	P95 (m)
wKNN (IDW)	18.24	27.25	10.14	46.08	65.34
wKNN C2F	18.18	30.05	7.16	57.23	75.35
NN Regression	19.79	29.03	11.06	54.70	71.93
CNN Regression	20.40	28.94	12.51	54.06	71.37
Ensemble (5 MLPs)	18.38	27.85	9.64	55.09	68.40





Key Findings: Localization

- **wKNN Coarse-to-Fine (C2F)** achieves best median error: 7.16 m
- **Ensemble methods** provide robust performance with stable tails
- **Classification fails** due to sparse grid (3,069 cells vs. $\sim$2,148 training samples)
- **Dominant error source:** Spatial ambiguity in NLOS shadow regions behind large facades
- **Fingerprinting robust** across different ray-tracing implementations (MATLAB validation confirms)

Channel Charting Methods

Goal: Produce low-dimensional representation preserving neighbourhood structure without position labels

- **PCA:** Linear principal-component projection (baseline)
- **t-SNE:** Non-linear stochastic neighbour embedding
  - Perplexity sweep: $\{5, 15, 30, 50\}$
  - Best: perplexity $= 30$
- **Autoencoder:** Symmetric encoder/decoder with 2-D bottleneck
- **UMAP:** Uniform manifold approximation and projection
  - Neighbourhood size sweep: $\{5, 10, 20, 30\}$
  - Best: $n_{\text{neighbors}} = 5$

Preprocessing: 50 PCA components for t-SNE and UMAP (noise reduction)

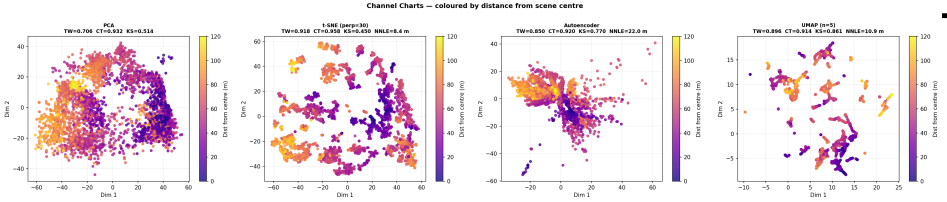
Channel Charting Evaluation Metrics

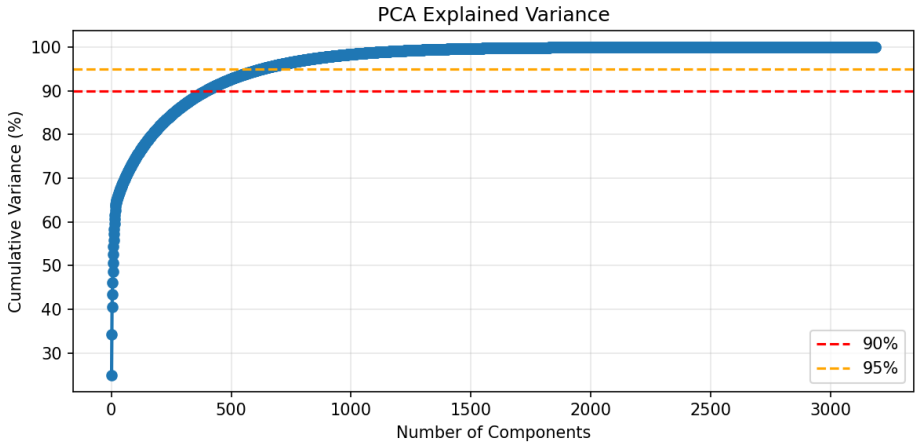
- **Trustworthiness (TW, ↑):** Penalises false neighbours (close in chart, far in RF space)
- **Continuity (CT, ↑):** Penalises tears (close in RF space, far in chart)
- **Kolmogorov–Smirnov Distance (KS, ↓):** Compares pairwise-distance distributions
  - KS $\to 0 =$ globally scale-preserving chart
- **NNLE (↓, m):** Nearest-neighbour localization error
  - Predicts UE position from chart's nearest neighbour
  - Ties chart quality to downstream task

Ideal Chart: TW $\approx$ CT ≈ 1, KS ≈ 0, NNLE $\sim$ grid spacing

Channel Charting Results

Method	TW ↑	CT ↑	KS ↓	NNLE (m) ↓
PCA	0.706	0.932	0.514	n/a
t-SNE (30)	**0.918**	**0.958**	**0.451**	**8.35**
Autoencoder	0.850	0.920	0.771	22.00
UMAP ($n = 5$)	0.896	0.914	0.861	10.92





Key Findings: Channel Charting

- **t-SNE dominates** across all metrics: TW=0.918, CT=0.958, KS=0.451
- **NNLE = 8.35 m** comparable to best supervised regressor without UE labels
- **UMAP trades** neighbourhood TW against global KS
- **Autoencoder limited** by MSE reconstruction objective (no metric preservation)
- **PCA baseline:** Preserves global structure but poorly preserves local topology
- **Cross-validation:** MATLAB dataset shows t-SNE robustness (TW=0.975, CT=0.945)

Summary of Results

- **Dataset:** MATLAB ray-tracing on Otaniemi campus with independent validation
- **Localization:** wKNN C2F achieves 7.16 m median error; ensemble methods robust
- **Channel Charting:** t-SNE produces geometrically faithful maps (TW=0.918, NNLE=8.35 m)
- **Insights:**
  - Fingerprinting robust across ray-tracing implementations
  - Regression superior to classification for dense spatial grids
  - Unsupervised charting competitive with supervised localization
- **Limitations:**
  - NLOS shadow regions cause ambiguity and inflated error tails
  - Limited training samples constrain deep learning approaches

Future Work

- **Enhanced Features:** Incorporate per-path AoA/AoD angles from ray-tracing output
- **Metric-Preserving Loss:** Replace MSE autoencoder with Siamese/triplet loss
- **Larger Scene:** Extend to full 77-building Otaniemi model (GPU acceleration needed)
- **Semi-Supervised Charting:** Use small subset of labelled anchors to regularise latent space
- **Practical Demonstration:** Beam-management or handover policy based on learned radio-space embedding